

AI-Driven Analyst Report Generation: A Multi-Agent LLM Approach to Investment Guidance

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Abstract

This paper presents a novel multi-agent framework leveraging large language models (LLMs) for automated financial analysis and investment report generation from earnings call transcripts. Traditional financial analysis struggles with the increasing volume of unstructured data, while single-agent LLM systems often suffer from hallucinations and difficulty managing complex analytical workflows. We propose a collaborative multi-agent system powered by *GPT* – 4.1 that mimics professional analyst team structures through role specialization. Our framework consists of three specialized agents namely, analyst, writer, and editor – that collaborate through structured workflows with tool support for financial data retrieval and sentiment analysis. Through extensive human evaluation with 176 participants on the Prolific platform. The system also scores highly on report quality dimensions, with particularly strong performance in readability (5.09/7) and logic (5.61/7). Our model’s average accuracy is 0.522. Our findings provide empirical evidence that role-based agent collaboration significantly enhances both the quality and practical utility of AI-generated financial analysis, offering a validated approach for augmenting human analysts in investment decision-making.

1 Introduction

Entity engagement and investment target prioritization have become increasingly critical for institutional investors navigating dynamic financial markets. This process heavily relies on financial analysis, which has traditionally depended on manual examination of structured data such as balance sheets (Loughran and McDonald, 2016). However, the exponential growth of unstructured data sources - including earnings calls, patent filings, and social media sentiment has rendered traditional approaches inefficient in capturing real-time market insights.

Early deep learning solutions attempted to automate parts of this process through sentiment analysis models and neural networks for risk forecasting (Loughran and McDonald, 2016). However, these approaches face significant challenges with data drift, where gradual or rapid changes in the input data distribution cause a degradation of model performance (Lu et al., 2018). Market dynamics such as regulatory changes, emerging sectors, and macroeconomic shocks alter data distributions, requiring costly retraining with substantial computational resources and labor intensive data labeling (Alzubaidi et al., 2021).

Recent advances in generative AI, particularly large language models (LLMs), offer a compelling alternative. Pre-trained on massive diverse corpora, LLMs can interpret complex contextual relationships without task-specific retraining. They excel at capturing subtle linguistic cues for nuanced sentiment analysis and have demonstrated strong performance in summarization, question answering, and market sentiment prediction (Yang et al., 2024). Domain-specialized models like BloombergGPT (Wu et al., 2023) and FinGPT (Yang et al., 2023) further demonstrate the benefits of adapting general-purpose models to financial text.

However, existing systems predominantly adopt single agent paradigms in which one LLM handles the entire analysis pipeline. While effective for narrow applications, these frameworks struggle with hallucinations causing factual inaccuracies (Kang and Liu, 2023) and incomplete coverage when tackling complex tasks such as investment reporting. Moreover, current financial AI frameworks lack realistic organizational modeling, failing to replicate the structured workflows and division of labor characteristic of professional analyst teams (Yu et al., 2023).

This paper addresses these limitations by exploring multi-agent LLM systems for financial anal-

ysis and automated investment report generation. We design and implement a collaborative agent framework powered by GPT-4.1 that analyzes financial textual data and produces well-informed investment recommendations. Our key contributions are as follows:

- A novel multi-agent framework that mimics professional analyst team structures through specialized role assignment
- Comprehensive evaluation comparing single-agent and multi-agent approaches on real earnings call transcripts
- Evidence that agent collaboration with iterative feedback significantly improves report quality and factual accuracy

2 Related Work

2.1 Explainability and Interpretability in Financial AI

The deployment of AI systems in financial contexts demands not only accuracy but also transparency and interpretability, particularly given regulatory requirements and the high-stakes nature of investment decisions. Recent work has comprehensively examined the landscape of explainable AI (XAI) in finance (Yeo et al., 2025), highlighting the critical need for systems that can provide faithful and interpretable explanations for their outputs.

The challenge of generating trustworthy explanations from LLMs has received considerable attention. Yeo et al. (2024) demonstrate through activation patching that natural language explanations from LLMs may not always faithfully represent their internal decision-making processes, raising important questions about the reliability of single-agent systems that lack verification mechanisms. This finding directly motivates our multi-agent approach, where the Editor agent serves as an additional layer of validation for the explanations and reasoning provided by other agents.

Interpretability concerns extend beyond individual model outputs to the reasoning processes themselves. Jie et al. (2024c) examine how interpretable reasoning explanations from prompted LLMs actually are, finding significant variability in explanation quality. Our multi-agent framework addresses this through role specialization: the Analyst agent provides data-grounded explanations, while the Editor ensures these explanations maintain logical consistency and clarity.

The extraction of interpretable rationales from financial text presents unique challenges. Jie et al. (2024b) propose semi-supervised approaches for extractive rationalization, which aligns with our Analyst agent’s function of identifying and extracting key financial metrics and insights from earnings transcripts. Similarly, Ong et al. (2023) introduce aspect-based sentiment analysis for explainable finance, demonstrating that decomposing financial analysis into specific aspects (similar to our agent specialization) improves both performance and interpretability.

The self-improvement capabilities of LLMs through knowledge detection (Jie et al., 2024a) suggest potential enhancements to our framework. While our current implementation uses fixed agent roles, future iterations could incorporate self-training mechanisms where agents learn from successful report generations to refine their specialized capabilities.

Finally, the challenge of structuring unstructured financial data, as addressed by Sun et al. (2024) in the context of ESG reports, parallels our task of converting free-form earnings call transcripts into structured investment reports. Their information extraction techniques could be integrated into our Analyst agent to enhance its ability to systematically extract and organize financial information.

These works collectively underscore that explainability and interpretability are not merely desirable features but essential requirements for financial AI systems. Our multi-agent framework inherently promotes explainability through its transparent workflow: each agent’s contribution is distinct and auditable, the iterative refinement process is traceable, and the use of external tools for validation provides grounded explanations for financial claims.

2.2 Generative AI in Financial Analysis

The financial sector has witnessed growing adoption of generative AI for analyzing complex textual documents. Recent studies demonstrate LLMs’ strong performance in summarization, question answering, and sentiment extraction from corporate earnings calls, 10-K filings, and analyst briefings (Yang et al., 2024; Chowdhery et al., 2023; Tournon et al., 2023). These models identify subtle language cues correlating with market movements, often outperforming human analysts in specific prediction scenarios (Hu et al., 2018).

Domain-adapted models further illustrate

the benefits of financial corpus training. BloombergGPT (Wu et al., 2023) achieves state-of-the-art performance in sentiment analysis and entity recognition, while FinGPT (Yang et al., 2023) demonstrates that open-source fine-tuned models can rival proprietary approaches on financial NLP benchmarks. However, hallucination remains a primary concern, where models fabricate plausible but inaccurate statements—particularly problematic in high-stakes settings where small factual errors can mislead investors (Kang and Liu, 2023). This motivates research into retrieval-augmented generation (RAG) that constrains LLM outputs with reliable external data (Gao et al., 2024).

2.3 Single-Agent AI Systems

AI agents extend LLMs into autonomous, goal-directed entities that operate more like human workers (Park et al., 2023; Sumers et al., 2023). These systems incorporate planning capabilities for multi-step actions, memory mechanisms for context maintenance, and tool use for accessing external resources (Parisi et al., 2022). This design enables agents to retrieve data through APIs, compute metrics, and generate fact-grounded reports rather than relying on speculative language (Yu et al., 2023).

Recent frameworks have operationalized these concepts successfully. GPT-Engineer demonstrated LLM-driven software generation (Qian et al., 2023), while Toolformer showed that LLMs can self-learn API usage (Schick et al., 2023). In finance, FinMem introduced layered memory architecture enhancing trading agents’ decision-making (Yu et al., 2023). However, single-agent systems face limitations as task complexity increases, remaining vulnerable to hallucinations and struggling with self-error correction without external feedback (Darwish et al., 2025).

2.4 Multi-Agent Systems and Role Specialization

To address single-agent limitations, recent research explores multi-agent systems where multiple LLMs interact under role-specific instructions (Hong et al., 2024; Qian et al., 2023). The premise draws on collective intelligence: specialized agent groups can outperform single generalist models when roles and workflows are well-defined (Zhang et al., 2024; Salve et al., 2024; Lu et al., 2023).

Several frameworks demonstrate this approach. MetaGPT (Hong et al., 2024) and ChatDev (Qian

et al., 2023) simulate software engineering teams with agents acting as product managers, coders, testers, and reviewers. Each role activates domain-relevant behavior in the underlying LLM, producing more reliable outputs than single-agent prompting. Research highlights the effectiveness of iterative feedback and debate structures where agents critique and refine each other’s work (Darwish et al., 2025).

Emerging literature applies multi-agent systems specifically to finance. Heterogeneous designs focusing on different error types improve financial sentiment analysis (Darwish et al., 2025). TradeMaster illustrates how reinforcement learning and multi-agent collaboration combine for quantitative trading (Sun et al., 2023). However, empirical exploration remains limited, with few systematic comparisons between single-agent and multi-agent approaches for financial analysis tasks.

3 Methodology

3.1 System Architecture Overview

We developed two prototype configurations to evaluate the effectiveness of multi-agent collaboration in financial analysis: (1) a single-agent baseline system and (2) a multi-agent collaborative system. Both configurations utilize GPT-4.1 as the underlying language model to ensure consistent capability across experiments, differing only in their approach to role specialization and agent interaction.

3.2 Single-Agent Configuration

The single-agent baseline employs a unified Writer agent that autonomously generates investment reports directly from earnings call transcripts. This agent consolidates all analytical and generative tasks—information extraction, analysis, and narrative synthesis—into a single pipeline.

The Writer agent operates with full autonomy, accessing conversational memory, planning capabilities, and registered tools including APIs for historical financial data retrieval and market sentiment analysis. Temperature is set to 0.5 to balance factual accuracy with natural language generation.

3.3 Multi-Agent Configuration

The multi-agent system employs three specialized agents collaborating through structured workflows:

Role-specific temperature settings optimize each agent’s function: Analyst and Editor operate at 0.2-0.3 for maximum accuracy, while Writer uses 0.7

Algorithm 1 Single-Agent Workflow

- 1: **Input:** Earnings call transcript T , Instructions I
 - 2: Writer receives (T, I) from Client
 - 3: Extract financial metrics from T
 - 4: Identify qualitative insights from T
 - 5: **if** external validation needed **then**
 - 6: Call `historicalFinancialData(ticker, year, quarter)`
 - 7: Call `analyzeMarketSentiment(ticker, year, quarter)`
 - 8: **end if**
 - 9: Synthesize findings into structured report R
 - 10: Client validates R against requirements
 - 11: **Output:** Investment report R
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Algorithm 2 Multi-Agent Workflow

- 1: **Input:** Earnings call transcript T , Instructions I
 - 2: Writer creates initial draft D_0 from T
 - 3: Analyst validates D_0 against external data:
 - 4: Extract metrics, calculate ratios
 - 5: Call external APIs for validation
 - 6: Generate structured feedback F_A
 - 7: Writer revises draft: $D_1 = \text{revise}(D_0, F_A)$
 - 8: **repeat**
 - 9: Editor reviews D_i for quality
 - 10: Check factual consistency
 - 11: Evaluate completeness and clarity
 - 12: Generate editorial feedback F_E
 - 13: Writer produces $D_{i+1} = \text{revise}(D_i, F_E)$
 - 14: **until** Editor approves or max iterations reached
 - 15: Client validates final draft D_n
 - 16: **Output:** Investment report D_n
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Agent	Responsibilities
Analyst	Extracts financial data, performs fact-checking, calculates ratios, conducts sentiment analysis using external tools
Writer	Drafts and revises investment reports incorporating Analyst data and Editor feedback, maintains professional tone and structure
Editor	Reviews drafts for accuracy, completeness, and readability, provides actionable feedback for revisions

Table 1: Multi-agent system roles and responsibilities

for natural language fluency. This configuration mimics real-world analyst teams where domain experts, writers, and editors collaborate iteratively.

3.4 Orchestration Framework

We implemented both systems using Microsoft AutoGen, an open-source framework for multi-agent LLM applications. AutoGen manages agent communication through its GroupChatManager, routing messages appropriately between agents. The framework handles message passing and turn-taking logic, simplifying implementation of iterative feedback loops.

3.5 Dataset and Preprocessing

We collected earnings call transcripts from the Alpha Vantage API for the "Magnificent 7" companies (Apple, Microsoft, Google, Amazon, Meta, Tesla, Nvidia), covering four quarters as of July 31, 2025. This provides a focused yet representative dataset of 28 transcripts from major technology firms.

Transcripts were converted from JSON to Markdown format, preserving speaker metadata (CEO, CFO, analysts) and structural sections (prepared remarks vs. Q&A). No truncation was required as transcript lengths fit within GPT-4.1’s context window. Each transcript was processed by both systems to generate paired outputs for comparison.

3.6 External Tools and APIs

Both configurations access two specialized tools:

- **historicalFinancialData(ticker, year, quarter):** Retrieves quarterly metrics (EPS, revenue, cash flow, balance sheet) from Alpha Vantage for year-over-year and quarter-over-quarter comparisons

316	• analyzeMarketSentiment(ticker, year, quarter): Collects news articles published within 30 days before the earnings call, ensuring realistic temporal constraints matching real analyst workflows	superior coverage of nuanced elements like risk factors and Q&A insights, likely due to the Editor’s feedback prompting more comprehensive revision.	360
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321	3.7 Evaluation Framework	4.3 Qualitative Analysis	363
322	We evaluate system outputs across multiple dimensions:	4.3.1 Report Structure and Professionalism	364
323		Multi-agent reports exhibited more consistent professional structure, typically following standard equity research format:	365
324	• Factual Accuracy: Verification of financial metrics against source transcripts and external data	• Executive summary with clear investment thesis	366
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326		• Detailed financial analysis with supporting data	368
327	• Completeness: Coverage of key topics including financial results, management commentary, and analyst Q&A	• Risk assessment and competitive positioning	369
328			370
329		• Forward-looking statements with justified projections	371
330	• Report Quality: Professional structure, clarity, and adherence to investment report standards		372
331		Single-agent reports, while coherent, often mixed analytical sections with summary content, lacking the clear delineation expected in professional reports.	373
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333	• Insight Generation: Depth of analysis beyond surface-level summarization		375
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335	Reports were anonymized as "Report A" and "Report B" to prevent evaluator bias during assessment.	4.3.2 Error Patterns	377
336		Single-agent systems showed characteristic error patterns:	378
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338	4 Experiments	• Conflation of quarterly and annual figures (12.3% of reports)	380
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340	4.1 Experimental Setup	• Unsupported extrapolations from limited data (8.9%)	382
341	We conducted experiments on 28 earnings call transcripts, generating paired reports from both single-agent and multi-agent configurations. Each system processed identical inputs with standardized parameters to ensure fair comparison. Token budgets were constrained to comparable ranges (approximately 2,000-3,000 tokens per report) to prevent length bias.	• Inconsistent sentiment interpretation (7.1%)	383
342			384
343		Multi-agent systems largely avoided these errors through the Analyst’s validation step, though occasionally introduced minor inconsistencies during Writer revisions (2.4% of reports).	385
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348	4.2 Quantitative Results	4.4 Computational Efficiency	390
349	4.2.1 Factual Accuracy Analysis	The multi-agent system required approximately 2.7× more computational resources due to multiple agent interactions and iterative refinement cycles. This trade-off between quality and efficiency represents a key consideration for deployment.	391
350	The multi-agent system demonstrated significantly higher accuracy across all metrics, with the most substantial improvements in ratio calculations and year-over-year comparisons where the Analyst agent’s specialized validation proved crucial.		392
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355	4.2.2 Completeness Evaluation	4.5 Ablation Studies	397
356	We assessed coverage of essential report components:	To understand individual agent contributions, we conducted ablation studies:	398
357		Results indicate that the Analyst’s external validation tools contribute most significantly to accuracy improvements, while the Editor’s feedback loop enhances overall quality and consistency.	399
358	While both systems consistently included financial highlights, the multi-agent approach showed		400
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5 Discussion

5.1 Key Findings

Our experiments demonstrate that multi-agent collaboration significantly enhances the quality of AI-generated financial analysis. The 16.4% improvement in factual accuracy and substantial gains in report completeness validate our hypothesis that role specialization and iterative refinement address key limitations of single-agent systems.

The Analyst agent’s integration of external data proved particularly valuable, reducing hallucinations and ensuring quantitative claims align with verifiable sources. The Editor’s quality control function, while contributing less to raw accuracy, substantially improved report professionalism and readability—critical factors for real-world deployment.

5.2 Practical Implications

For financial institutions considering AI augmentation, our findings suggest multi-agent systems offer superior reliability for high-stakes analysis. The increased computational cost (approximately 2.8× higher) may be justified for investment-grade reports where accuracy is paramount. However, single-agent systems remain viable for rapid preliminary analysis or lower-stakes summarization tasks.

The modular nature of multi-agent systems also enables targeted improvements. Organizations could improve specific agents (e.g., adding specialized risk assessment agents) without restructuring the entire system, a significant advantage for iterative development.

5.3 Comparison with Human Analysts

Although our system generates comprehensive reports, several distinctions from human analysis remain:

- Limited ability to incorporate non-textual market signals
- Absence of industry-specific intuition and experience
- Difficulty identifying subtle management communication patterns
- Inability to conduct primary research or channel checks

These limitations position our system as an augmentation tool rather than replacement for human analysts, best suited for initial report generation that analysts can refine with domain expertise.

6 Limitations

Several limitations constrain the generalizability of our findings:

Dataset Scope: Evaluation on only "Magnificent 7" technology companies may not reflect performance across diverse sectors with different reporting conventions.

Temporal Constraints: Using transcripts from a single year limits assessment of system adaptability to varying market conditions.

Model Dependency: Exclusive use of GPT-4.1 prevents comparison with other LLMs that might exhibit different collaboration dynamics.

Evaluation Metrics: Lack of standardized benchmarks for investment report quality necessitated manual evaluation, potentially introducing subjective bias.

Real-world Validation: Absence of actual trading performance data prevents assessment of report utility for investment decisions.

7 Conclusion

This paper presented a novel multi-agent framework for automated financial analysis that significantly outperforms single-agent baselines in generating accurate, comprehensive investment reports. Through role specialization mimicking professional analyst teams, our system demonstrates that structured LLM collaboration can address critical limitations of monolithic approaches, particularly regarding factual accuracy and analytical depth.

Our contributions advance the field of financial NLP by providing empirical evidence for multi-agent superiority in complex analytical tasks and offering a practical framework for augmenting human analysts. As financial markets generate increasingly complex data streams, such AI-augmented analysis tools become essential for maintaining competitive advantage.

Future work should explore dynamic agent team composition, cross-domain generalization, integration with quantitative trading systems, and real-time market data incorporation. Additionally, developing standardized benchmarks for financial report generation would enable more systematic

comparison of approaches. Ultimately, we envision multi-agent systems becoming integral to institutional investment processes, enhancing human decision-making while maintaining the judgment and intuition that remain uniquely human contributions to financial analysis.

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References

Laith Alzubaidi, Jinglan Zhang, Amjad J Humaidi, Ayad Al-Dujaili, Ye Duan, Omran Al-Shamma, José Santamaría, Mohammed A Fadhel, Muthana Al-Amidie, and Laith Farhan. 2021. Review of deep learning: Concepts, cnn architectures, challenges, applications, future directions. *Journal of Big Data*, 8(1):1–74.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, and 1 others. 2023. Palm: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113.

Ahmed M Darwish, Ehab A Rashed, and Gasser Khoriba. 2025. Mitigating llm hallucinations using a multi-agent framework. *Information*, 16(7):517.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. 2024. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.

Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiaowu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, and 1 others. 2024. Metagpt: Meta programming for a multi-agent collaborative framework. *International Conference on Learning Representations, ICLR*.

Ziniu Hu, Weiqing Liu, Jiang Bian, Xuanzhe Liu, and Tie-Yan Liu. 2018. Listening to chaotic whispers: A deep learning framework for news-oriented stock trend prediction. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, pages 261–269.

Yeo Wei Jie, Teddy Ferdinan, Przemyslaw Kazienko, Ranjan Satapathy, and Erik Cambria. 2024a. Self-training large language models through knowledge detection. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15033–15045.

Yeo Wei Jie, Ranjan Satapathy, and Erik Cambria. 2024b. Plausible extractive rationalization through semi-supervised entailment signal. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 5182–5192.

Yeo Wei Jie, Ranjan Satapathy, Rick Goh, and Erik Cambria. 2024c. How interpretable are reasoning explanations from prompting large language models? In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2148–2164.

Haoqiang Kang and Xiao-Yang Liu. 2023. Deficiency of large language models in finance: An empirical examination of hallucination. *arXiv preprint arXiv:2311.15548*.

Tim Loughran and Bill McDonald. 2016. Textual analysis in accounting and finance: A survey. *Journal of accounting research*, 54(4):1187–1230.

Jie Lu, Anjin Liu, Fan Dong, Feng Gu, Joao Gama, and Guangquan Zhang. 2018. Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, 31(12):2346–2363.

Pan Lu, Baolin Peng, Hao Cheng, Michel Galley, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, and Jianfeng Gao. 2023. Chameleon: Plug-and-play compositional reasoning with large language models. *arXiv preprint arXiv:2304.09842*.

Keane Ong, Wihan Van Der Heever, Ranjan Satapathy, Erik Cambria, and Gianmarco Mengaldo. 2023. Finxabsa: Explainable finance through aspect-based sentiment analysis. In *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 773–782. IEEE.

Aaron Parisi, Yao Zhao, and Noah Fiedel. 2022. Talm: Tool augmented language models. *arXiv preprint arXiv:2205.12255*.

Joon Sung Park, Joseph C O’Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. *arXiv preprint arXiv:2304.03442*.

Chen Qian, Xin Cong, Chenglong Yang, Weize Chen, Yusheng Su, Juyuan Xu, Zhiyuan Liu, and Maosong Sun. 2023. Communicative agents for software development. *arXiv preprint arXiv:2307.07924*.

Aniruddha Salve, Saurabh Attar, Mandar Deshmukh, Shashank Shivpuje, and Ujjwal A Mitra. 2024. A collaborative multi-agent approach to retrieval-augmented generation across diverse data. *arXiv preprint arXiv:2412.05838*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36:68539–68551.

Theodore R Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L Griffiths. 2023. Cognitive architectures for language agents. *arXiv preprint arXiv:2309.02427*.

- Shuo Sun, Molei Qin, Wentao Zhang, Haochong Xia, Chuqiao Zong, Jie Ying, Yonggang Xie, Lingxuan Zhao, Xinrun Wang, and Bo An. 2023. Trademaster: A holistic quantitative trading platform empowered by reinforcement learning. *Advances in Neural Information Processing Systems*, 36:59047–59061.
- Zounachuan Sun, Ranjan Satapathy, Daixue Guo, Bo Li, Xinyuan Liu, Yangchen Zhang, Cheng-Ann Tan, Ricardo Shirota Filho, and Rick Siow Mong Goh. 2024. Information extraction: Unstructured to structured for esg reports. In *2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 487–495. IEEE.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, and 1 others. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
- Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kam-badur, David Rosenberg, and Gideon Mann. 2023. Bloomberggpt: A large language model for finance. *arXiv preprint arXiv:2303.17564*.
- Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. Fingpt: Open-source financial large language models. *arXiv preprint arXiv:2306.06031*.
- Jingfeng Yang, Hongye Jin, Ruixiang Tang, Xiaotian Han, Qizhang Feng, Haoming Jiang, Shaochen Zhong, Bing Yin, and Xia Hu. 2024. Harnessing the power of llms in practice: A survey on chatgpt and beyond. *ACM Transactions on Knowledge Discovery from Data*, 18(6):1–32.
- Wei Jie Yeo, Ranjan Satapathy, and Erik Cambria. 2024. Towards faithful natural language explanations: A study using activation patching in large language models. *arXiv preprint arXiv:2410.14155*.
- Wei Jie Yeo, Wihan Van Der Heever, Rui Mao, Erik Cambria, Ranjan Satapathy, and Gianmarco Mengaldo. 2025. A comprehensive review on financial explainable ai. *Artificial Intelligence Review*, 58(6):1–49.
- Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W Suchow, and Khaldoun Khashanah. 2023. Finmem: A performance-enhanced llm trading agent with layered memory and character design. *arXiv preprint arXiv:2311.13743*.
- Wentao Zhang, Lingxuan Shen, Jingwei Yang, Hao-ran Zhu, Zhangyang Chen, Dongyan Sun, Jiazheng Zhang, Xiaodong Li, and Yuxuan Zhang. 2024. A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist. *arXiv preprint arXiv:2402.18485*.

sectionExample Appendix

This is an appendix.